

MUHAMMAD FACHRI SURYAPRAJA

Jakarta, Indonesia | [linkedin.com/in/fachrisuryapraja](https://www.linkedin.com/in/fachrisuryapraja) | +62 878-2288-8377 | fachrisuryapraja02@gmail.com

EDUCATION

Institut Teknologi Bandung

Bandung, Indonesia

BS, Water Resources Engineering and Management (GPA: 3.63/4.00)

August 2018 – October 2022

- **Achievements:** Received Dean's List for Academic Achievement: Academic Year 2019, 2020, 2021, and 2022, Ganesha Karya Awardee, Karya Salemba Empat (KSE) Scholarship
- **Activities and Societies:** Society of Renewable Energy-ITB (Community Service), Water Project-ITB, Himasda-ITB (Water Resources Engineering Student Association)
- **Final Thesis:** Flood Control Planning and Critical Riverbank Management In Kapuas River, Kapuas Hulu Regency, West Kalimantan Province

WORK EXPERIENCE

Associate - Business Development & Hydropower Engineer

Jakarta, Indonesia

PT. United Tractors Tbk

Jan 2025 – Present

- Responsible for conducting feasibility studies, planning and design, project management, and environmental evaluations to ensure the development and execution of hydro projects are technically sound, environmentally compliant, and cost-effective.
- Develop and implement strategic initiatives to expand the company's presence in the hydropower sector by identifying new markets, project opportunities, and business partnerships.
- Provide technical expertise to support business development activities, including interpreting hydrological and geological data, conducting site assessments, and optimizing project designs for efficiency and sustainability.
- Evaluate the technical feasibility and economic viability of potential hydroelectric projects by analyzing water resource availability, site conditions, and environmental impacts.

Hydrologist

Jakarta, Indonesia

PT. Indokoei International (Nippon Koei Group)

Mar 2023 – Dec 2024

- Conduct hydrological studies to assess the availability and distribution of water resources in the project area
- Calculated annual energy that produced by the hydropower plant estimated based on hydrological analysis and determined turbine types which are suitable for the location.
- Assisted civil engineer in designing the main component of hydropower in terms of hydraulic analysis such as weir, headrace, headpond, sediment trap, penstock, etc

Junior Water Resources Engineering

Bandung, Indonesia

PT. Suropati Hidro Energi

Sep 2022 – Feb 2023

- Conducted hydrological studies to assess water availability, analyzing precipitation, runoff, and groundwater data
- Assisted in the development and implementation of hydraulic models to simulate water flow and assess the impact of infrastructure on river systems.
- Supported senior engineers in the preparation of feasibility studies, providing insights into water resource assessments and potential environmental impacts

Research Graduate Assistant

Bandung, Indonesia

Institut Teknologi Bandung

May 2022 – February 2023

- Programmed and developed software to assist decision-making regarding the operation of dams and watersheds knowledge namely Indonesia River Simulation Decision Support System (IRIS-DSS) with the R program
- Constructed a comprehensive study on Siak River Basin Flood Management Subproject in Riau Province
- Conducted a thematic study on the development of Tegal Luar Reservoir in the framework of fulfilling clean water demand for citizens in Bandung Regency, West Java Province

ORGANIZATIONAL EXPERIENCE

Water Resources Engineering Student Association (Himasda-ITB)

Bandung, Indonesia

President

April 2021 – March 2022

- Coordinated & determined the direction for all lines of activities in the Himasda-ITB's student association through 9 departments, 4 divisions & almost 150 staff.
- Developed, implemented, and evaluated the overall organization's strategy.
- Collaborated and networking with other internal and external organizations to achieve the vision of Himasda-ITB

AWARDS, TECHNICAL SKILLS, AND INTERESTS

Awards:

- *1st Winner of International Dam Innovation Contest (IDIC) 2022* with a paper entitled : "Sediment Bypass Tunnel Innovation In Sustainable Dam Planning as an Effort to Reduce Sedimentation Rate & Minimize Environmental Impact" | Diponegoro University
- *1st Winner of UNESCO Water Resilience Challenge (UWRC) 2021* with a master plan entitled : "Alan Sembalun: Bring Sembalun Back to Nature" | UNESCO and The Water Agency (TWA)
- *1st Winner of Indonesia Water Building National Competition (KBAI) 2020* with a paper entitled : "Innovation of Smart Irrigation Based on Renewable Energy to Support Modernization of Irrigation in Indonesia" | Brawijaya University

Technical Skills: SWMM (Drainage Modelling) | HEC-RAS (River & Flood Modelling) | HEC-HMS (Hydrology Modelling) | ArcGIS (Geoprocessing Analysis) | AutoCAD (Technical Drawing)

Interests: Integrated Water Resources Management (IWRM), Economic in Water, Irrigation and Drainage System, Water Disaster Mitigation, Dam Engineering, Hydropower, Leadership, Public Speaking